

Course Name: Physics II Lab

Course Code: Phy 4242

Experiment No.: 06

Experiment Name: Verification of Superposition
Theorem

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Date of Submission: 07/03/2023

Objective:

- i) To be familiar with the superposition theorem and know its applicability.
- ii) To verify this theorem which is an analytical technique of determining currents in a circuit with more than one source.

Theory:

It states that, "In any, bilateral network consisting of more than one source of emf, the current through or voltage across any element is equal to the algebraic sum of the currents or voltages produced independently by each source."

While the current due to a particular source of EMF is being found, the other emf sources are rendered inactive and if any branch element is in series with those sources that remain intact.

Circuit Diagram :

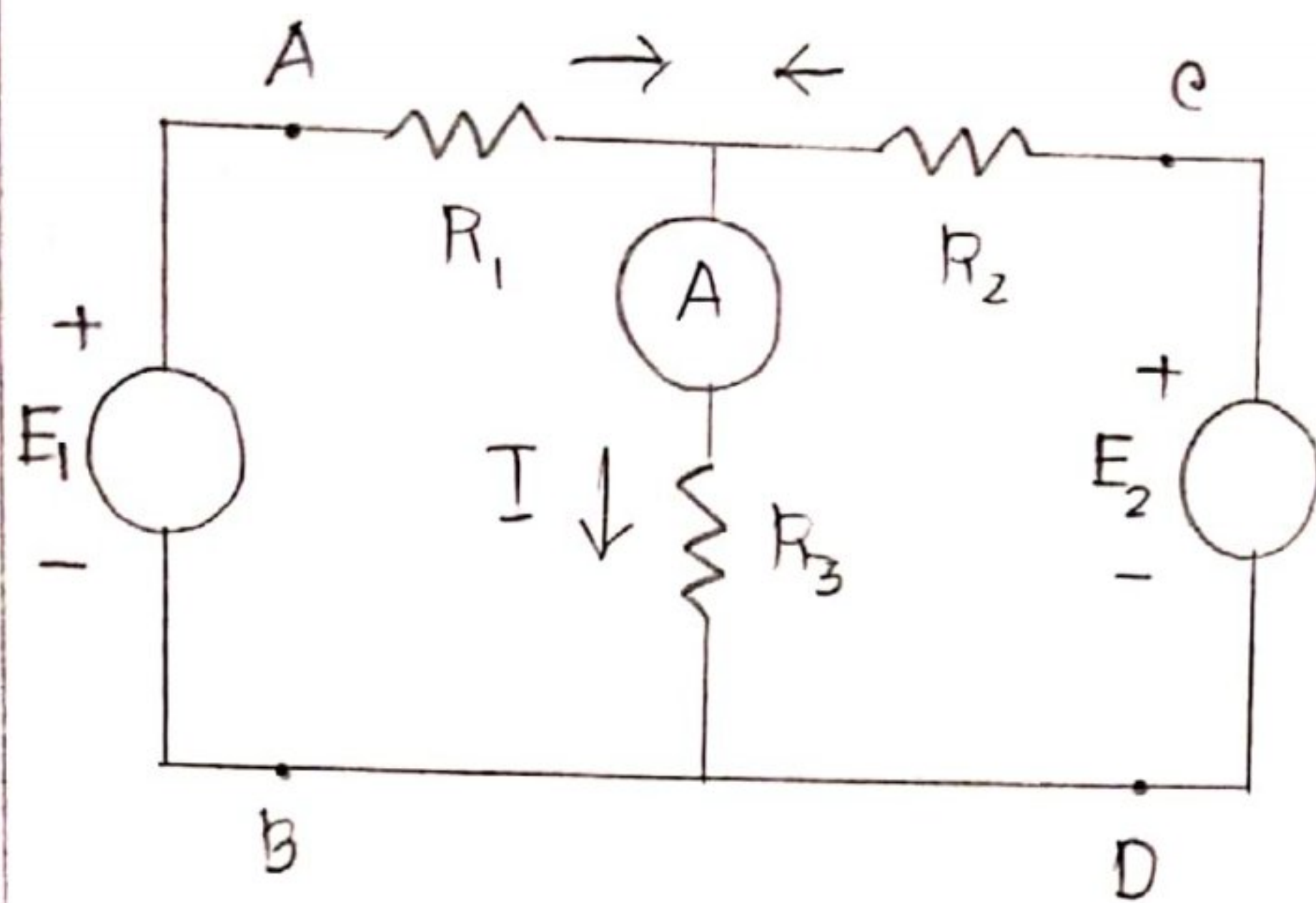


Fig. a

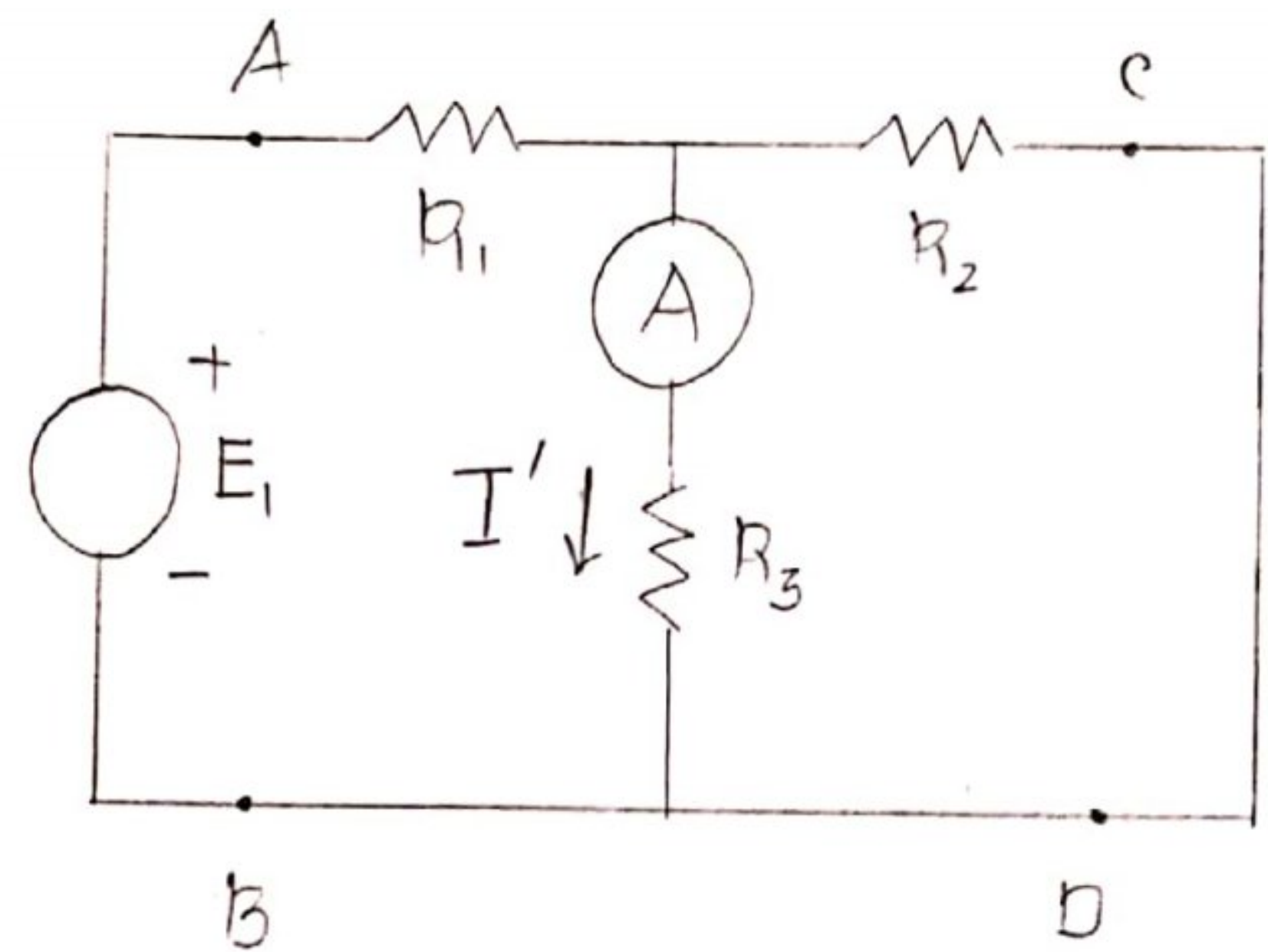


Fig. b

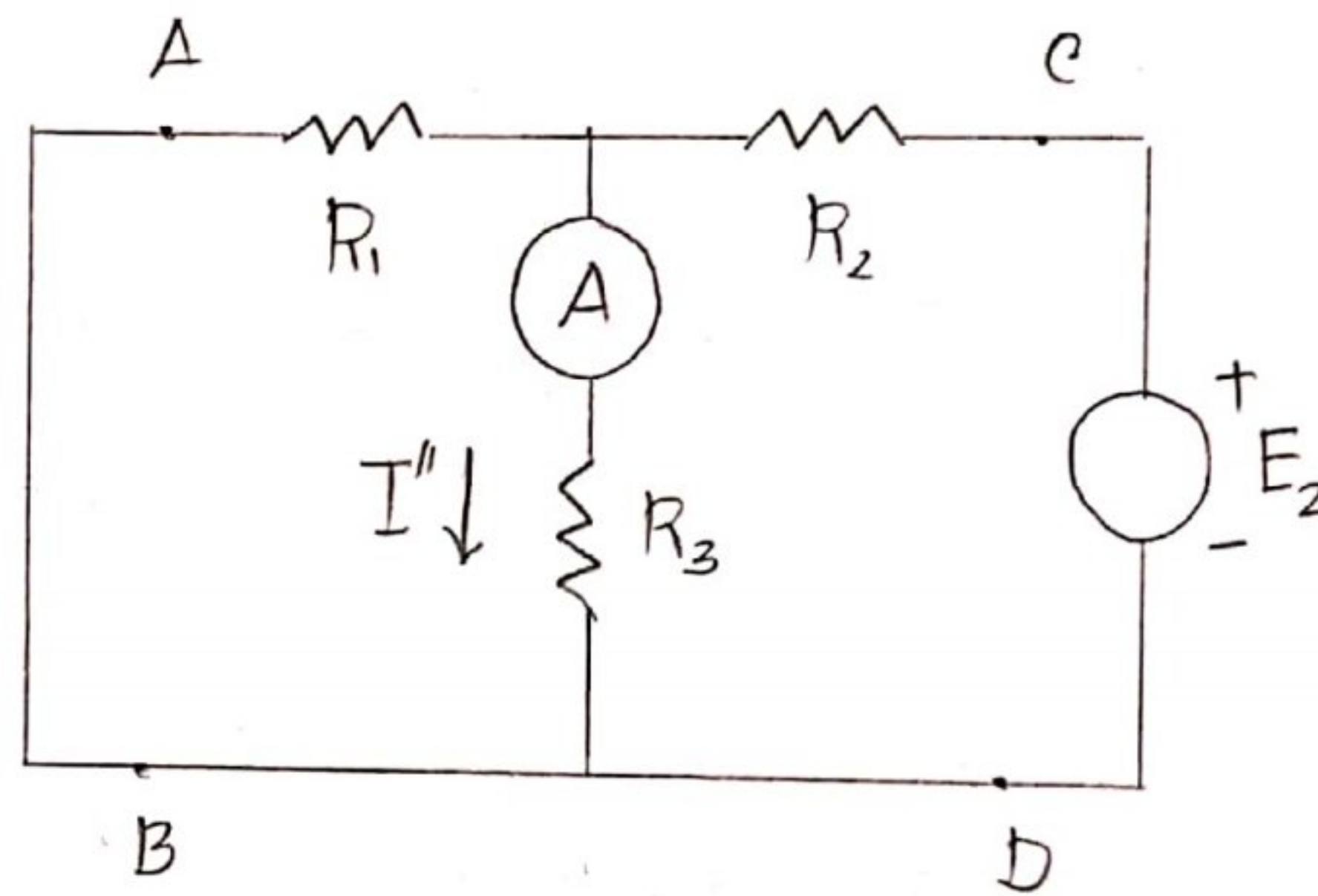


Fig. c

Data Collection :

For first observation, measured resistances,

$$R_1 = 2142 \Omega$$

$$R_2 = 3.27 \text{ k}\Omega$$

$$R_3 = 550 \Omega$$

From Fig. b,

$$R_{eq} = (R_2 \parallel R_3) + R_1 = 2612.8115 \Omega$$

$$I_1 = \frac{5}{R_{eq}} = 1.9136 \times 10^{-3} A$$

$$\begin{aligned} \therefore I'_1 &= \frac{3270}{3270+550} \times 1.9136 \text{ mA} \\ &= 1.63 \text{ mA} \end{aligned}$$

From Fig. c,

$$R_{eq} = (R_1 \parallel R_3) + R_2 = 3707.63 \Omega$$

$$I_2 = \frac{-15}{R_{eq}} = -4.0457 \text{ mA}$$

$$\begin{aligned} \therefore I'' &= \frac{2142}{2142+550} \times (-4.0457 \text{ mA}) \\ &= -3.2191 \text{ mA} \end{aligned}$$

$$\begin{aligned} \therefore I &= I'_1 + I'' \\ &= -1.5891 \text{ mA} \end{aligned}$$

So, theoretically, for the first observation with

2142 Ω resistance, $I = -1.5891 \text{ mA}$

For 2nd observation,

From figure b,

$$R_{eq} = (R_2 \parallel R_3) + R_1 = 7130.8115 \Omega$$

$$I_1 = \frac{5}{R_{eq}} = 0.7012 \text{ mA}$$

$$\therefore I' = \frac{3270}{3270 + 550} \times 0.7012 \text{ mA} = 0.6 \text{ mA}$$

From figure c,

$$R_{eq} = (R_1 \parallel R_3) + R_2 = 3778.044 \Omega$$

$$I_2 = \frac{-15}{R_{eq}} = -3.97 \text{ mA}$$

$$\therefore I'' = \frac{6660}{6660 + 550} \times (-3.97 \text{ mA}) = -3.667 \text{ mA}$$

$$\therefore I = I' + I''$$

$$= -3.067 \text{ mA}$$

Table:

Obs. No.	Values of R_1, R_2, R_3 (ohms)	I (mA)	I' (mA)	I'' (mA)	$I = I' + I''$ (mA)
1	2142, 3270, 550	-1.60	1.63	-3.30	-1.67
2	6660, 3270, 550	-3.18	0.55	-3.69	-3.14

Data Analysis:

For first observation,

Theoretically, $I = -1.5891 \text{ mA}$

Practically, $I = -1.67 \text{ mA}$

For second observation,

Theoretically, $I = -3.067 \text{ mA}$

Practically, $I = -3.14 \text{ mA}$

Result:

The experiment is successful in verifying Superposition theorem. We find that by using $I = I' + I''$

both practically and theoretically we get almost

the same value of I .

Discussion and Q&A:

- 1) Each resistors were measured carefully.
- 2) The components were connected correctly following the diagram.
- 3) There are some discrepancies between the theoretical and practical values of the measured current. This is due to various human errors or the equipment not being calibrated properly.